

PAGASA SYNOP Validator

Operational Checking Ruleset

Version 0.14.3 - Separated Ruleset

Document status	Operational review guide
Coding form	WMO FM 12 SYNOP (AAXX), with PAGASA practices
Validator input	One complete observation at a time
Published build	Ruleset v0.14.3-separated
Primary references	WMO Manual on Codes, Volume I.1 (2019); PAGASA Amended Guidelines on Surface Synoptic Observation (adopted 01 June 2023)

Purpose

This document describes the checks implemented in the PAGASA SYNOP Validator. It is intended for operational review, regression testing, and controlled ruleset maintenance. It does not replace the official WMO code manual, PAGASA observing instructions, or observer judgment.

Important

A result of 'No coded errors detected' means only that the report passed the checks currently implemented. It is not a certification that every meteorological observation or coding decision is correct.

1. Scope, inputs, and result levels

The validator reads the complete bulletin or AAXX observation, separates Sections 1, 3, and 5, decodes selected groups, and cross-checks related groups. Supporting inputs are deliberately limited to keep the page practical.

Input	Use
Complete SYNOP code	Required. Includes AAXX, YYGGiw, station identifier, coded groups, and preferably '='.
MSLP 3 hours earlier	Checks the exact sign category and ppp magnitude in 5app.
MSLP 24 hours earlier	Used at 00 and 12 UTC to check the Section 3 58/59 group.
Rainfall occurred checkbox	Cross-checks rainfall groups, iR, and present/past weather without requiring a previous coded observation.

Finding levels

Level	Meaning
Error	The coded report conflicts with a required format, schedule, arithmetic relationship, or implemented operational rule.
Warning	The report is possible but needs observational review, supporting information, or confirmation.
Information	A section or optional check is unavailable.
OK	No error or warning was produced by the implemented rules.

Correction policy

- Suggestions name the decoded element where possible, such as CM=2 (Altostratus opacus or Nimbostratus), rather than showing only a code figure.
- Automatic suggestions never replace checking the original observation.
- A 0.1 hPa mismatch in pressure-history comparisons is an error; values must agree exactly at coded precision.

2. Bulletin structure and mandatory syntax

Element	Implemented check
Bulletin heading	Checks the normal TTAAii, CCCC, and YYGGgg patterns when a bulletin header is present.
AAXX	Required before the observation time and station identifier.
YYGGiw	Five digits; day 01-31, hour 00-23, and iw limited to 0, 1, 3, or 4.
Iliii	Five-digit station identifier.
Section 1 mandatory groups	iRixhVV and Nddff must follow the station identifier.
Section markers	333 starts Section 3; 555 starts Section 5.
Group width	Every coded data group must contain five figures or permitted solidi.
End marker	Missing '=' produces a warning.
Duplicates	Repeated groups in Sections 1, 3, or 5 are reported.
Section 3 order	Standard groups are checked for increasing indicator order.

Key format checks

iRixhVV is checked for allowed precipitation, station-operation, cloud-base, and visibility figures. Nddff is checked for total-cloud amount, wind direction 00-36 or 99, and wind speed figures.

Example

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SMPH20 RPLC 171200 AAXX 17121 98327 11460 62001 ... 333 ... =
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3. Pressure and temperature checks

Pressure values are decoded to tenths of a hectopascal. Comparisons use integer tenths so floating-point rounding cannot hide a 0.1 hPa discrepancy.

Rule	Check and result
4PPPP plausibility	MSLP must be within the configured operational range 870.0-1085.0 hPa.
5appp magnitude	Absolute current-minus-3-hour pressure change must equal ppp exactly.
5appp sign category	Positive: a=0-3; negative: a=5-8; zero: a=0,4,5. The shape value still requires trace review.
24-hour pressure	At 00/12 UTC, 58 is positive/zero and 59 is negative; magnitude must match exactly.
Zero 24-hour change	59000 is rejected; zero must use 58000.
Air temperature	A valid 1snTTT group is required.
Dew point	A valid 2snTdTd group is required; it must not exceed air temperature.
Section 3 extremes	00 UTC requires minimum-temperature 2-group; 12 UTC requires maximum-temperature 1-group.
Extremes vs current	Minimum cannot exceed current temperature; maximum cannot be below current temperature.

Not checkable from the coded report alone

The PAGASA instruction to round raw temperatures conventionally to 0.1 C, as well as broken-thermometer and psychrometric procedures, requires the original readings and is outside this one-code validator.

4. Rainfall, weather, and visibility

Topic	Implemented rule
Main/intermediate schedule	Main hours: 00, 06, 12, 18 UTC. Intermediate hours: 03, 09, 15, 21 UTC.
iR location	Checks required presence and placement of 6RRRtR in Sections 1 and/or 3.
tR schedule	Expected tR: 4 at 00; 1 at 06/12/18; 7 at 03/09/15/21 UTC.
Rainfall checkbox	If rainfall occurred, requires a measurable/trace rainfall representation and reviews ww/W1/W2 support.
Weather group vs ix	Checks required or conflicting 7wwW1W2 presence.
Unexplained low visibility	VV at or below 5 km warns when ww gives no visibility-reducing phenomenon.
Precipitation/cloud form	Drizzle expects Stratus; rain expects stratified clouds; rain showers expect Cu or Cb.

Visibility ranges used

Phenomenon/intensity	Expected horizontal visibility
Light precipitation	>= 9 km
Moderate precipitation	2 km to < 9 km
Heavy precipitation	<= 2 km
Present weather 40 (fog at a distance)	<= 2 km
Light haze	6 km to < 10 km
Moderate haze	1 km to < 6 km
Thick haze	< 1 km
Light fog	500 m to < 1 km
Moderate fog	200 m to < 500 m
Thick fog	< 200 m

Severity convention

When the reported visibility is lower than the suggested precipitation range, the validator gives a warning. When it is higher than the permitted range, the validator gives an error.

5. Cloud rules

Cloud checks combine Nddff, h in iRixhVV, 8NhCLCMCH, and the Section 3 8NsChshs groups.

PAGASA 1-3-5 selection rule

- Report the lowest individual layer when Ns is 1 okta or more.
- Report the next higher individual layer when Ns is 3 oktas or more.
- Report the next higher individual layer when Ns is 5 oktas or more.
- Cumulonimbus is always reported if it was not selected by the preceding criteria.
- Maximum: three selected layers plus one additional Cb group.
- Order groups from lower to higher cloud base.

Independent layer amounts - critical v0.14 correction

Each individual layer amount is estimated as if no other clouds existed. Layers can overlap. The validator therefore does not add Section 3 Ns values to derive Nh. It checks that an individual low-cloud Ns does not exceed Nh.

Official PAGASA example

85800 333 83820 84630

Interpretation: Nh=5; 3 oktas Cumulus at 600 m; 4 oktas Stratocumulus at 900 m. The individual values total 7 but the code is valid because the layers overlap.

Sky state	Required treatment
N=0	Omit the main 8NhCLCMCH group and all Section 3 individual cloud groups.
N=9	Omit the main cloud group and report 89/hshs, where hshs is vertical visibility.
N=/	Omit the main cloud group and Section 3 individual cloud groups.

6. Cloud type, height, and reportable-cloud matching

Cross-check	Implemented interpretation
Cloud level	Low below 2,000 m; middle 2,000-6,000 m; high above 6,000 m.
Lowest cloud base h	Compared only with the lowest reported low-cloud layer, not every low layer.
CL=8	May support both C=8 Cumulus and C=6 Stratocumulus.
CL=7	May support C=7 Stratus and C=8 Cumulus fractus.
CM=2	Supports C=4 Altostratus and C=5 Nimbostratus.
CM=7	Supports Altocumulus with Altostratus or Nimbostratus.
Missing cloud family	A reportable low/middle/high genus conflicts when the corresponding CL/CM/CH indicates no cloud.
Wrong indicator	A Section 3 group beginning with 3 may be suggested as 8NsChshs when amount, genus, and height are otherwise plausible.
56DDD	Direction figures are checked against no-cloud and unobservable-cloud indicators.

Known correction example

83108 333 56909 83820 87360

CM=0 means no middle cloud, but 87360 reports C=3 Altocumulus. With CH=8 and a 9,000 m base, the intended correction may be 87280 after confirming the observation.

CB with other low clouds

The presence of Cumulonimbus does not mean no other low clouds exist. Nh is checked against all qualifying reported low-cloud layers without requiring their amounts to add exactly.

7. CB, gustiness, Section 5, and schedules

Group/rule	Implemented check
949CD	When Cb is reported, requires at least one valid 949 group. C=4 isolated; C=5 numerous; D=0-9 direction. Multiple groups are allowed.
CB reportable layer	CL indicating Cb requires an 8Ns9hshs reportable Cb layer; other low-cloud layers remain possible.
904tt / 907tt	Must be followed immediately by 911ff and 915dd.
911ff / 915dd	Orphan speed or direction groups are rejected; dd must be 00-36 or 99.
Section 5 2-group	At day 1 00 UTC, decoded as the national six-hour rainfall for 18-24 UTC.
6//// RRRRR	Extended previous-month rainfall form, permitted at day 1 00 UTC and intended for totals exceeding 1,000 mm.
Monthly 6RRRR	Section 5 monthly-rainfall groups at other times are errors.

Observer criteria not derivable from AAXX

- Gustiness requires the wind-trace criteria: at least three marked positive peaks, one reaching 8 m/s; peak-lull variation at least 5 m/s; and average interval at most 20 seconds.
- Thunderstorm intensity and lightning frequency require observer/timeline information.
- Precipitation-rate intensity requires amount and duration measurements.
- Continuous/intermittent precipitation and thunderstorm-period rules require an event timeline.

8. Feedback, maintenance, and limitations

The site uses a separated architecture so operational rules can be reviewed without mixing them with page controls.

File	Responsibility
ruleset-config.js	Editable schedules, thresholds, version, pressure limits, cloud levels, CB codes, and monthly-rain threshold.
ruleset.js	Parsing, decoding, meteorological relationships, errors, warnings, and corrections.
app.js	Buttons, pressure inputs, rainfall checkbox, screen rendering, and feedback email draft.
index.html	Page structure and script loading order.
styles.css	Visual design.
tests/ruleset-regression.test.mjs	Known correct and incorrect observations used to prevent regressions.

Report checker issue - v0.14.2

The button opens an editable email draft addressed to renieragas@gmail.com. The draft contains the entered code, version, summary, and finding titles. If the device has no default email handler, the page attempts to copy the report for manual pasting. The site does not automatically transmit or store the observation.

Operational limitations

- The validator cannot see the original sky, instrument trace, raw measurements, or observer worksheet.
- Warnings may be appropriate when multiple meteorological interpretations are possible.
- National practices not yet supplied or encoded remain outside the ruleset.
- Every rule change should include a known correct case, a known erroneous case, and a regression-test update.

9. Regression reference and version summary

Reference case	Expected purpose
84908 333 81915 83820 87280 94945	Correct Cb plus another low cloud; Nh=4 from observed low-cloud cover, without assuming Cb is the only low cloud.
85800 333 83820 84630	Official overlapping-layer example; must not produce an Ns-sum error.
N=9 with 333 89/00	Correct obscured-sky vertical-visibility form.
83108 ... 87360	Detects middle-cloud type conflict and gives a named correction.
Exact pressure history	Passes when 5app and 58/59 match; a 0.1 hPa difference is an error.
904tt/907tt 911ff 915dd	Checks complete gust sequence and direction coding.
Day 1 00 UTC 555 20005 6//// 12602	Recognizes six-hour and extended previous-month rainfall forms.

Version changes relevant to this ruleset

Version	Main change
v0.12	Expanded operational checker from accumulated testing notes.
v0.13	Separated settings, meteorological logic, and webpage controls.
v0.13.1	Fixed browser script-scope collision and added browser-load regression guard.
v0.14	Aligned cloud overlap, CL/CM mapping, and N=0/9// handling with the PAGASA guideline.
v0.14.1	Added email feedback draft.
v0.14.2	Improved mail-link compatibility, shortened the draft, and added clipboard fallback.
v0.14.3	Heavy precipitation and present weather 40 accept visibility of 2 km or less.

End of ruleset

Review this document together with the coded examples, current source files, official PAGASA observing instructions, and the WMO Manual on Codes.